

Power Plants and the Electrical Grid

In Section 13.3, the historical competition between the use of AC and DC electricity was discussed. Efficiency was one of the main reasons why AC electricity won over DC. But why is AC more efficient? To answer that question, we must first examine the transmission of electrical energy along a conductor.

Transmission Efficiency and Current

Generators at today's large-scale power plants can produce huge amounts of power. A significant amount of this power can be lost due to thermal energy losses as large currents pass through the transmission wires. To determine how much power is lost in a transmission wire, we can use the power equation $P = VI$. To solve for lost power, we can derive a new power equation using Ohm's law to substitute for V :

$$P = VI$$

$$P = (IR)(I)$$

$$P = I^2R$$

This equation can be used to calculate the amount of power lost in the wire due to thermal energy losses.

Assume that a generator produces 300 MW (3×10^8 W) of power at a current of 30 kA, which travels through a transmission wire with a resistance of 0.1Ω . Using the new power equation we find

$$\begin{aligned} P &= I^2R \\ &= (30 \text{ kA})^2(0.1 \Omega) \\ &= (30\,000 \text{ A})^2(0.1 \Omega) \\ &= 9 \times 10^7 \text{ W} \\ P &= 90 \text{ MW} \end{aligned}$$

So 90 MW of power is transformed to unusable thermal energy. This represents a loss of 30 %.

Note that the lost power is proportional to the square of the current, so if we could lower the current going through the wire, there would be much less power lost. Fortunately, a transformer can lower the current, increase the voltage, and keep the same power. By using a transformer at a power plant we step up the voltage. Suppose that we increase the voltage to 100 kV. This lowers the current to 3 kA. Repeating the calculation, we find

$$\begin{aligned} P &= I^2R \\ &= (3 \text{ kA})^2(0.1 \Omega) \\ &= (3000 \text{ A})^2(0.1 \Omega) \\ &= 9 \times 10^5 \text{ W} \\ P &= 0.9 \text{ MW} \end{aligned}$$

This represents a loss of 0.3 %, which is a significant improvement. This is the main reason why we generate AC electricity at power plants. Transformers will only work with AC electricity. Without the transformer, the amount of power lost in transmission would be impractical.

The Electrical Power Grid

The electrical power grid is a giant circuit composed of many parallel circuits fed by the electrical energy of a number of power plants. The grid transmits AC power using transformers that step up and step down the voltage where necessary (**Figure 1**).

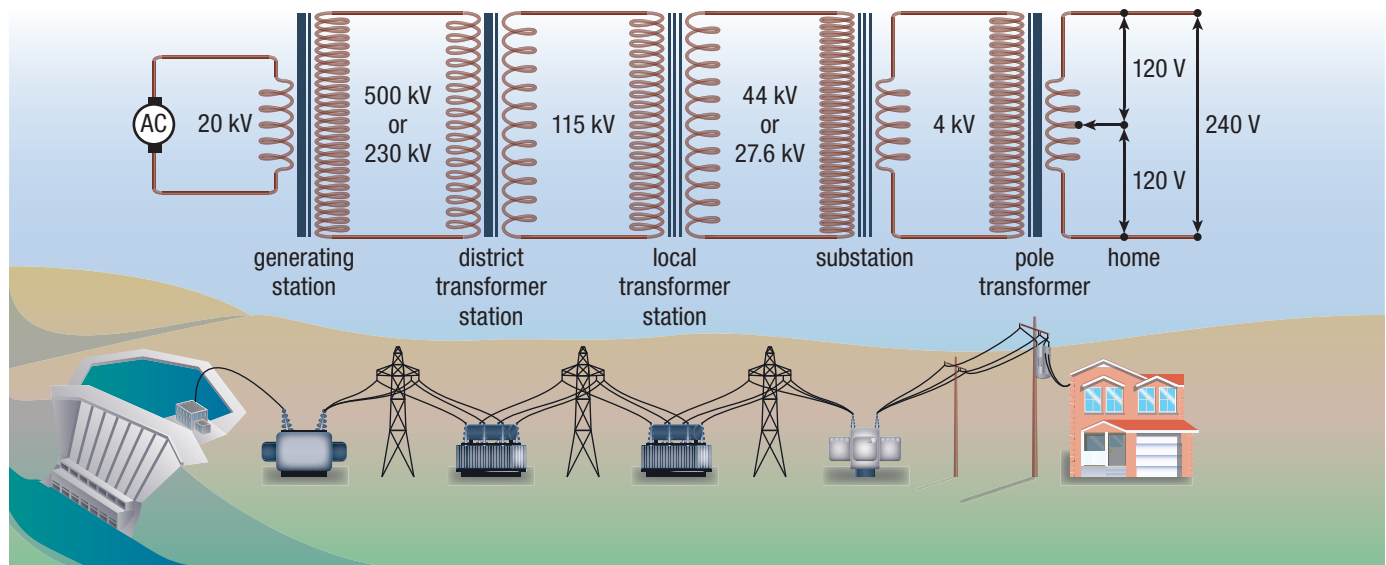


Figure 1 The step-up and step-down transformers used in a typical electrical grid

In Figure 1, the generator produces 20 kV of AC, which is immediately increased to 230 kV or higher to minimize energy loss. The electricity is then sent along power transmission lines suspended high above the ground supported by towers. If the voltages were higher, then the electricity could discharge through the air and into the ground. The electricity is gradually decreased in voltage at a district transformer station, a local transformer station, a substation (**Figure 2**), and then a pole or ground transformer in your neighbourhood (**Figure 3**).

The electrical power grid is monitored, and energy is fed into the grid on demand. If more energy is needed, and there is the capacity, then more is fed in. Power plants only generate the amount of electricity that is needed because electrical energy cannot easily be stored. If more is generated than needed, it is sold to other electrical grids farther away. If more electrical energy is needed and we do not have the capacity, then we purchase it from other grids at a higher cost. On some summer days in Ontario, when demand is high, we may use more electrical energy than the power plants are capable of generating. So we purchase electricity from the United States in order to meet the increased need.

The electrical grid needs maintenance. As the grid ages, continued repair and replacement are needed and the costs are passed along to the consumer. If you look at your electricity bill, you will find that there are costs listed for the amount of electricity used and its delivery. The delivery fee is collected to maintain the grid.

Commercial AC Generators

The generators used in power plants contain multiple coils and armatures. The field magnets are not permanent magnets because it is difficult to make a strong enough magnet. Also, permanent magnets lose their magnetism over time because of the strong magnetic fields in the coils. So, instead, electromagnets are used. To increase the strength of an electromagnet, you increase the current. Where does the electrical energy come from to power the electromagnets? In some cases, it comes from the AC generator itself. In other cases, the AC generator has a DC generator that uses



Figure 2 Transformers at this substation step down the voltage so that it is low enough to be transmitted to neighbourhoods.



Figure 3 This residential transformer steps down the voltage to 240 V for use in your home.

permanent magnets. The DC generator uses a source of energy (such as falling water) to generate electrical energy, which is then used to power the electromagnets of the AC generator. The AC generator uses a source of energy (such as falling water) along with the DC generator to generate AC electricity. **Figure 4** shows cross-sections of a large-scale generator and a hydroelectric power plant.

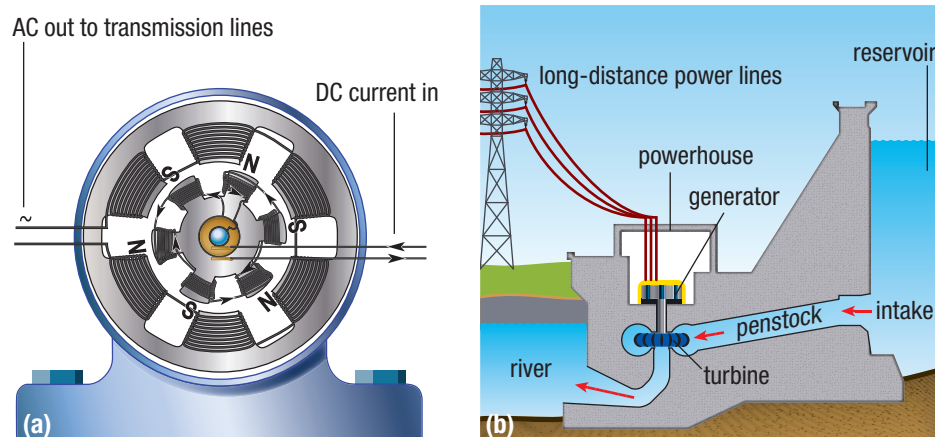


Figure 4 (a) Cross-section of a large generator (b) Cross-section of a hydroelectric power plant

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The rotation speed of a generator must be managed to maintain the desired frequency of 60 Hz. As electricity demand increases, more current is drawn into the grid and it becomes more difficult to turn the generator. You can increase the turning force of the turbine rotating the generator shaft. If this cannot be done, you can decrease the strength of the electromagnets inside the generator. This lowers the voltage. It is for this reason that the voltages continually fluctuate a small amount throughout the day. The voltages are required to be relatively constant, however, and can only fluctuate within a regulated amount. 

13.6 Summary

- Transmission of AC electricity requires the use of transformers to minimize losses.
- Step-up transformers at the power plant are used to increase the voltage and decrease the current for transmission.
- Step-down transformers are used throughout the grid to bring voltages down to levels that can be used in homes.
- Commercial generators have multiple armatures and coils using electromagnets to generate AC electricity.

13.6 Questions

1. Describe the main reason why AC power generation was chosen over DC power generation. K/U C
2. Determine the power loss in each of the following. Express your answer as a percent. T/I
 - (a) A 200 MW power plant delivers a current of 2 kA in a 10 Ω wire.
 - (b) A 200 MW power plant delivers a current of 200 A in a 10 Ω wire.
 - (c) A 10 MW wind turbine delivers a current of 3000 A in a 0.50 Ω wire.
3. Why is electrical energy generated on demand? K/U C
4. What is the difference between the electrical generators you learned about in Section 13.4 and the commercial generators discussed in this section? C A
5. Look at Figure 1 on page 611. The pole transformer has a secondary coil with a connection point in the middle of the coil as well as at the ends. Why do you suppose that is the case? K/U
6. Commercial generators were described as using electromagnets which are sometimes powered by the generator itself. Would it be possible to just let the generator power itself without the need for an external energy source? Explain your answer. K/U